

Results

01 • FACTORIAL ANOVA

main effects + interaction of 2+ factors

Table 1. Cell descriptives

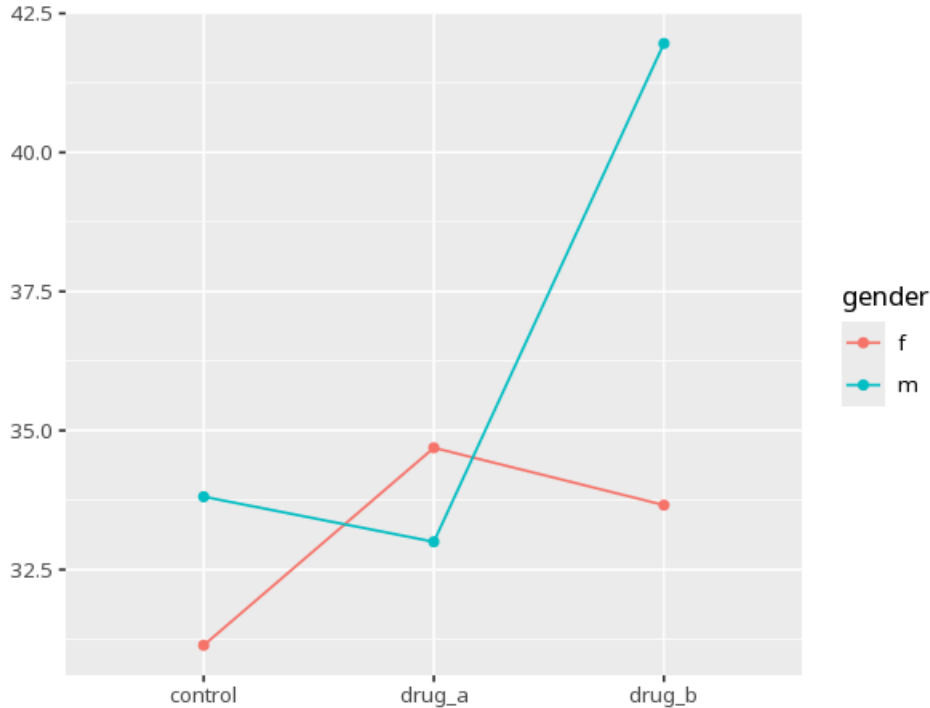
Factor A × B	N	M	SD
control × f	10	31.14	8.22
drug_a × f	10	34.69	7.22
drug_b × f	10	33.66	8.27
control × m	10	33.81	6.79
drug_a × m	10	33.00	6.86
drug_b × m	10	41.96	4.65

Table 2. ANOVA (main effects + interaction)

Source	SS	df	MS	F	p	partial η^2 [95% CI]
group	307.07	2	153.53	3.04	.056	0.10 [0.00, 1.00]
gender	143.53	1	143.53	2.84	.098	0.05 [0.00, 1.00]
group × gender	250.84	2	125.42	2.48	.093	0.08 [0.00, 1.00]

assumption checks: Levene's & normality of residuals; post-hoc / simple-effects table when an effect is significant. (Levene $F=0.51$, $p=.766$ · Shapiro $W=0.97$, $p=.184$)

Figure. Interaction



How to read this test

Each factor's F/p is its main effect; the A×B row tests whether the effect of one factor depends on the other. A significant interaction usually takes priority — read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones — use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold (α) is 0.05.

APA template: A two-way ANOVA gave A×B interaction $F(2,54)=2.48$, $p = .093$, partial $\eta^2=.08$ [.00, 1.00].

After export · optional feedback — an anonymous, optional satisfaction survey — opens a short Google Form in a new tab

[How did it go? — Share feedback →](#)